Price wedge anomaly signals: construction log

September 12, 2026

How to read this

One entry per characteristic. *Agreement* is the mean correlation, across the ten deciles and 462 formation months, between our decile capitalisation weights and the replication package's own (`datas_all_charP`). It tests the characteristic and the sort alone, independently of the 241-month tracking and the discount factor. Above 0.85 the construction is treated as verified. *Error* is our price wedge less the paper's Table 1 value, in percentage points.

Rejected alternatives records what was tried and did not work, so the same ground is not covered twice.

Summary

Signal	Group	Agreement	Long err.	Short err.	Status
OA	intangibles	0.963	-1.53	0.15	verified
AOA	intangibles	0.971	-1.15	0.05	verified
DP	value	0.924	1.46	-4.16	verified
ATO	profitability	0.974	-0.86	0.16	verified
RNA	profitability	0.978	0.60	-0.30	verified
dSOUT	investment	0.992	-0.53	-0.68	verified
dPIA	investment	0.943	0.23	-0.87	verified
TNOVR	trading frictions	0.993	0.09	-0.10	verified
SUV	trading frictions	0.982	-0.47	-0.11	verified
DTO	trading frictions	0.985	-0.69	-0.59	verified
aBEME	value	0.896	-0.20	-0.85	verified
aSAT	profitability	0.896	-0.05	1.09	verified
aSIZE	trading frictions	0.987	-1.40	-0.22	verified
A2ME	value	0.972	3.57	-0.13	verified
AT	trading frictions	0.988	-0.02	0.11	verified
BEME	value	0.977	1.24	-0.70	verified
C2A	value	0.963	-0.06	-0.20	verified
C2D	value	0.959	-0.48	-2.03	verified
CAT	profitability	0.977	1.23	-0.32	verified
D2P	value	0.964	0.47	-0.36	verified
dCEQ	investment	0.951	-2.79	-1.51	verified
dGS	profitability	0.966	-0.69	-0.25	verified

Signal	Group	Agreement	Long err.	Short err.	Status
dSO	value	0.966	-0.40	-0.53	verified
E2P	value	0.978	3.00	-0.41	verified
EPS	value	0.977	0.82	-3.88	verified
IDIOV	trading frictions	0.955	-0.21	-0.99	verified
I2A	investment	0.971	0.01	-1.30	verified
IPM	profitability	0.962	-0.78	-1.43	verified
IVC	investment	0.970	-0.98	0.26	verified
SIZE	trading frictions	0.997	0.09	-0.21	verified
NOA	investment	0.967	-1.12	-0.26	verified
OL	intangibles	0.959	1.35	-0.16	verified
BETA_d	trading frictions	0.990	-0.01	-0.11	verified
PCM	profitability	0.977	-0.44	-1.57	verified
PM	profitability	0.973	-0.61	-1.53	verified
PROF	profitability	0.978	-0.53	-0.29	verified
Q	value	0.978	0.91	-0.21	verified
MAXRET	trading frictions	0.984	-0.22	-1.16	verified
ROA	profitability	0.965	-0.06	0.53	verified
ROC	profitability	0.972	-0.95	0.24	verified
ROE	profitability	0.972	-0.41	-2.69	verified
ROIC	profitability	0.964	0.07	-0.23	verified
R_12_2	past returns	0.979	0.17	0.29	verified
R_12_7	past returns	0.984	-0.13	-0.43	verified
R_6_2	past returns	0.978	-0.02	-0.91	verified
R_2_1	past returns	0.986	-0.35	-0.45	verified
R_36_13	past returns	0.988	-0.95	-0.40	verified
S2C	value	0.968	0.15	-0.92	verified
S2P	value	0.979	5.01	-0.62	verified
SG	value	0.963	0.12	-0.82	verified
SAT	profitability	0.971	1.65	-0.20	verified
SPREAD	trading frictions	0.985	-0.31	-0.07	verified
sdTURN	trading frictions	0.991	0.03	-0.06	verified
sdDVOL	trading frictions	0.989	-0.42	-0.11	verified
TAN	intangibles	0.978	-1.35	-1.38	verified
RETVOL	trading frictions	0.988	-0.14	-0.32	verified
aPM	profitability	0.778	-1.40	-0.71	unresolved

Where the paper and its data disagree

Graded deliberately. Only category A claims something is wrong. B and C are documentation that cannot be built from; D concerns the reported results rather than the signals; E records things that look like discrepancies and are not, so they are not re-opened.

A. An error in the signal itself

The construction contradicts the formula the paper states, the wording of its own appendix, and the paper it cites, and the difference changes what the signal measures.

OA, AOA

Documented: Appendix Table A.1: "Operating Accruals: Changes in non-cash working capital minus depreciation (dp), all scaled by lagged total assets." Freyberger, Neuhierl and Weber word it identically and cite Sloan (1996), whose equation 1 subtracts depreciation.

Implemented: Depreciation sits inside the current-liabilities bracket, so expanding gives "+ dp": accruals are computed net of everything except depreciation, which is added.

Evidence: Netting depreciation out agrees with the package's own published decile weights at 0.286; adding it gives 0.963. Decile one's share of market capitalisation comes out at 5.29% against a published 5.30%, and the long-leg wedge error falls from +39.9 percentage points to -1.5. AOA, the absolute value, goes from 0.041 to 0.971.

Why it matters: OA and AOA are not operating accruals. Depreciation is large and persistent for capital-intensive firms, so those firms are sorted toward the high-accrual leg by their depreciation rather than away from it. The sign of the depreciation term is the whole content of Sloan's measure.

Note: Chen and Zimmermann's PctAcc.do carries the identical bracket placement while their Accruals.do does not, which suggests both inherited the same transcription rather than arriving at it independently.

SUV

Documented: Appendix Table A.1, citing Garfinkel (2009): "Difference between actual volume and predicted volume. Predicted volume is from a regression of previous month's daily volume on a constant and the absolute values of positive and negative previous month's returns. Unexplained volume is standardized by the standard deviation of the residuals from the regression."

Implemented: The regression is estimated on the current month's trading days, and the signal is the last trading day's standardised residual from it – one day's volume surprise, not the month's.

Evidence: Against the paper's own firm-month panel (t_by_n/SUV.mat, shared by Andrea Tamoni in September 2026) the last day's within-month residual matches at Spearman 0.9999 with a ratio of 1.0000; agreement with the published decile weights 0.982. The documented reading, and every month-level variant of it, agrees with the panel at between -0.001 and -0.04, and with the published weights at 0.07.

Why it matters: The decile a firm is sorted into is decided by its volume on one trading day relative to that month's own fit. This is the signature of a daily series collapsed to monthly by taking the last observation rather than the month's value, and the same signature appears in DTO. As published, SUV is not Garfinkel's standardised unexplained volume.

DTO

Documented: Appendix Table A.1, citing Garfinkel (2009): "Daily volume (vol) to shares outstanding (shrout) minus the daily market turnover and detrended by the 180 trading day median", with Nasdaq volume scaled down by 50% before 1997 and 38% after.

Implemented: The last trading day's turnover less the mean of its own previous 180 trading days. No market turnover is subtracted, Nasdaq volume is not scaled, and the detrending uses a mean, not a median.

Evidence: Spearman 0.998 against the paper's panel (Pearson 0.997); agreement with the published decile weights 0.985. The 180-day median instead of the mean gives 0.888 against the panel; subtracting the market's turnover or scaling Nasdaq volume lowers it further (the documented form, sampled on the last day, 0.56-0.64; averaged over the month, 0.18-0.31).

Why it matters: Without the market adjustment the signal is a firm's own one-day turnover shock rather than abnormal turnover in Garfinkel's sense, and as with SUV a single day stands in for the month.

B. The appendix cannot be built from

The implementation is defensible, often the only thing possible, but somebody working from Table A.1 alone would get a materially different series. Agreement scores below are what the literal reading achieves against what the data does.

SPREAD

Documented: "Average daily bid-ask spread in the previous month", citing Chung and Zhang (2014), whose estimator is the quoted spread.

Implemented: Each trading day contributes its high-low range, 2(high-low)/(high+low), if it has one and its quoted spread otherwise; the month is the mean over those days, with no minimum number of days.

Evidence: literal reading 0.02, range alone 0.94. Built day by day, every firm-month in 1975, 1985, 1995 and 2005 agrees with the paper's panel within 1% and none the panel has is missing.

Why it matters: Not a defect in the data – CRSP carries quotes for 9% of daily rows in the 1960s and 53% in the 1980s, so a quoted spread cannot be computed over this sample at all – but the citation points at an estimator the sample cannot support, and the substitution is unstated.

sdDVOL

Documented: "Standard deviation of residuals from a regression of daily volume (vol) on a constant", i.e. the dispersion of daily share volume in levels.

Implemented: The standard deviation of the log of daily *dollar* volume over the days that have volume; zero-volume days are left out.

Evidence: share volume in levels 0.06, log share volume 0.63, log dollar volume 0.87. Against the paper's panel, including zero-volume days as ln(1 + 0) disagrees by a factor of 1.6 to 5 for every firm-month that has one (27% of them); leaving them out matches.

Why it matters: Three undocumented choices at once: the units, the logarithm, and the exclusion of zero-volume days. In levels the measure captures firm size rather than variability.

dSOUT

Documented: "Annual % change in shares outstanding (shrout)."

Implemented: Growth in shrout multiplied by CRSP's cumulative share adjustment factor.

Evidence: literal reading 0.51, implemented 0.97

Why it matters: Taken literally the signal reads every two-for-one split as a 100% issuance. This is the largest gap between a literal reading and the data of any signal outside category A.

ROIC

Documented: "...to the sum of common equity (ceq), total liabilities (lt), and cash and short-term investments (che)."

Implemented: Cash is subtracted from invested capital, not added.

Evidence: as worded 0.64, as implemented 0.93

Why it matters: Here the wording is wrong and the implementation is right – invested capital is capital tied up in the business, net of cash – so this is a typo in the appendix rather than a problem with the data.

BETA_d

Documented: "The sum of the regression coefficients of daily excess returns on market excess returns and the lag of market excess returns." No estimation window is given.

Implemented: Estimated over the firm's last 249 trading days, the market's lag taken on the firm's own rows.

Evidence: one month 0.35, six 0.62, twelve 0.94, twenty-four 0.63 on the published weights. Against the paper's panel, 249 of the firm's days puts 84% of firm-months within 1% with a rank correlation of 0.9996; 250 days 47%, calendar twelve months 24%.

Why it matters: The window is the difference between a usable signal and noise, and it is not stated. Separately, FNW's own entry under this name describes the Frazzini-Pedersen beta – a correlation times a volatility ratio – which is a different estimator from the Dimson sum A.1 describes.

DP

Documented: "Sum of monthly dividend over the last 12 months to last month's price (prc)."

Implemented: Dividends in dollars over the *current* month's market capitalisation.

Evidence: per share over price 0.84, dollars over market cap 0.92

Why it matters: Minor, but per share the share count drifts through the year and the two are not the same quantity.

TAN

Documented: The formula is given; the missing-data rule is not.

Implemented: Every component required non-missing.

Evidence: zero-filling 0.48, requiring all 0.96

Why it matters: Zero-filling reads as "this firm holds no tangible assets" and sends firms with incomplete filings to the bottom decile.

TNOVR

Documented: "Volume (vol) over shares outstanding (shrout)."

Implemented: The previous month's volume over the current month's shares outstanding.

Evidence: as documented 0.71, with the one-month lag 0.978; Spearman 1.0000 against the paper's panel

Why it matters: A one-month misalignment of numerator and denominator. Harmless for most firms, but a split or an issuance between the two months changes the share count without the volume, so the signal halves or jumps for exactly the firms whose turnover is in question.

aBEME, aPM, aSAT, aSIZE

Documented: "Industry level is defined as the Fama-French 48 industries."

Implemented: Twenty-seven industry groups that are not Fama-French 48, recovered from the panels themselves: within a month the value less its adjusted value is one constant per industry, and grouping SIC codes by that constant month after month gives 27 groups that never contradict one another. The SIC code is CRSP's name-history code (msenames), which reproduces the panel's own siccd for 99.5% of firm-months.

Evidence: Fama-French 48 with the same inputs agrees with the published decile weights at 0.51 (aBEME), 0.45 (aSAT), 0.87 (aSIZE); the recovered scheme at 0.92, 0.91, 0.99. One recovered group holds 410 SIC codes spanning five FF48 industries (food, agriculture, textiles, clothing, construction); another 281 codes across building materials, machinery, steel, paper and household goods.

Why it matters: Coarser than FF48 across manufacturing and matching no published classification we tried (FF 5, 10, 12, 17, 30, 38, 48, 49; one- to three-digit SIC). It could be recovered only because the panels expose the cell means; from the appendix alone these four signals cannot be rebuilt.

C. The appendix drops a qualifier its own source insists on

Table A.1 is a condensation of Freyberger, Neuhierl and Weber. In a few places the condensation loses something that changes the answer.

A2ME, E2P, S2P, BEME, SIZE

Documented: FNW state the market capitalisation timing explicitly – "as of December t-1" for A2ME, E2P and BEME, "as of December" for S2P, "of the previous month" for SIZE. Table A.1 says only "market capitalization (prc x shrout)" with no timing at all.

Implemented: The current month's market capitalisation.

Evidence: contemporaneous 0.95-0.99 across the five, December of the prior year 0.61-0.97

Why it matters: The data does not follow FNW here, and A.1 is silent rather than wrong. But anyone building these from FNW, as the citation invites, would use December and land a long way off – Q falls from 0.98 to 0.61, E2P from 0.97 to 0.69.

Q

Documented: FNW write "the market value of equity (SHROUT times PRC) minus cash and short-term investments (CEQ)" – CEQ is common equity, not cash.

Implemented: Common equity, which is what CEQ means.

Why it matters: A typo in FNW that Table A.1 silently corrects. Recorded because it runs the other way.

D. Issues in the reported results rather than the signals

Found while replicating the paper's own tables and out-of-sample tests.

Table of characteristics-to-wedge mappings

Documented: Profitability alone is reported with an R-squared of exactly 0.00, alongside a non-zero coefficient of +0.13.

Implemented: We obtain 0.18 on the same 114 extreme portfolios.

Evidence: every other entry in that table replicates within a few points: PC3 0.705 against 0.76, PC6 0.752 against 0.77, FF5 0.704 against 0.74, BEME 0.550 against 0.67

Why it matters: A zero R-squared with a non-zero slope is hard to produce and is probably a reporting slip.

aPM, dGS

Documented: Each is signed so its long-short earns a positive one-month alpha, and reported with a long and a short leg.

Implemented: Their one-month alphas in our data are 0.00012 and 0.0013 a month.

Evidence: our sign convention agrees with the package for 55 of 57 characteristics; these are the two that differ

Why it matters: For aPM especially the alpha is indistinguishable from zero, so which leg is "long" is arbitrary and the published long and short wedges for it could swap without anything being wrong.

The out-of-sample firm-level correlation of 0.98

Documented: Firm wedges from a pre-1998 mapping correlate 0.98 with full-sample firm wedges, offered as evidence the approach is robust out of sample.

Implemented: Both series are linear functions of the same three principal components of the same firm characteristics, differing only in coefficients.

Evidence: Refitting the mapping to portfolio wedges with HALF the spread gives firm wedges correlating 1.0000 with the correct ones; fitting to wedges plus noise of equal variance gives 0.9976.

Why it matters: A mapping that gets every magnitude wrong by a factor of two passes this test perfectly. It establishes that the coefficient vector is stable, which is a real if modest claim, but it never compares a prediction against a realised wedge.

The out-of-sample portfolio correlation of 0.92

Documented: First-half portfolio wedges correlate 0.92 with full-sample wedges.

Implemented: The first half is one half of the full sample, so the two overlap by construction.

Evidence: For two halves correlating rho, each correlates with the whole at about sqrt((1+rho)/2). At the paper's footnoted rho of 0.77 that is 0.94; at ours, 0.58, it is 0.89 and we observe 0.95.

Why it matters: The 0.92 is close to implied by the 0.77 in the footnote. The footnote is the informative number, and it is the one we cannot reproduce.

The mapping R-squared of 0.76

Documented: Three principal components explain 0.76 of portfolio price wedges.

Implemented: Computed on the 114 extreme deciles only.

Evidence: on all 570 portfolios the same mapping gives 0.543 in sample and 0.411 leaving a characteristic family out; the direct rank regression gives 0.822 and 0.749 on the same portfolios

Why it matters: Restricting to the extremes flatters three principal components specifically – the gap to a direct regression is 0.09 on the extremes and 0.31 on all ten deciles. Most firms sit in the middle deciles.

Time-split out-of-sample tests

Documented: Wedges are recomputed on subsamples of formation dates.

Implemented: The market price of risk is calibrated once on the full sample and carried over.

Evidence: Carried over, early cohorts average -37 percentage points and late ones +33, a seventy-point flip that is purely the difference in realised market returns across the cut. Re-solving it within each half – 3.9 to 4.9 early against 2.4 to 2.9 late – the flip disappears and root mean squared error falls from 37-77 points to 10-15.

Why it matters: Correlations are location-invariant so the paper's reported correlations are unaffected. It does matter for the Panel B investment-q regressions, which use out-of-sample firm wedges as a regressor in levels.

E. Looks like a discrepancy, is not

Recorded so these are not re-opened. Each cost real time to resolve.

The five momentum sorts

Documented: "Cumulative return from 12 months to 2 months ago."

Implemented: The panel stores, on row t, the return over months t-11 to t-1 – one month later than the name. The sort then lags every characteristic by a further month, so the signal a portfolio formed at t actually sorts on is the return from t-12 to t-2, exactly as documented.

Evidence: storing the literal window instead drops agreement from 0.97 to 0.65, and for R_2_1 from 0.97 to 0.11

Why it matters: NOT an error. The economics is right and the stated definition is what gets used; only the storage convention is undocumented. It matters enormously for anyone rebuilding the panel and not at all for the paper's results.

R_12_2 stored as a gross return

Documented: A cumulative return.

Implemented: Stored as 1 + r rather than r; the median difference against ours is exactly 1.0000.

Why it matters: A units convention. Ranks are untouched, so no sort is affected.

Market capitalisation units

Implemented: Stored in millions in the package and in thousands in CRSP, a factor of exactly 1000.

Why it matters: Value weighting is scale-invariant, so nothing is affected.

The panel's row alignment

Implemented: MainPart1.m slices the panel from one row *before* July 1926, so its rows are offset by a month relative to the obvious reading.

Evidence: Comparing without noticing makes contemporaneous market equity look like a 3% match and lagged market equity a 91% match, which is backwards. Implementing the lagged version accordingly made the sorts worse – 50 verified characteristics down to 48, BEME from 0.97 to 0.84 – which is what exposed the error.

Why it matters: A trap for anyone comparing against the shipped panel, not a problem with the paper.

Construction, signal by signal

OA — Operating accruals

Group intangibles **Source** Sloan (1996), per Appendix Table A.1 **Frequency** annual **Agreement** 0.963

Inputs. `act, che, lct, dlc, txp, dp, at`

Formula.

> OA = (d act - d che - (d lct - d dlc - d txp - dp)) / at[t-1]
> where d txp treats a missing value as zero.
> Note the placement of dp: it sits INSIDE the bracket, so expanding gives
> OA = d act - d che - d lct + d dlc + d txp + dp, i.e. depreciation is
> ADDED to accruals.

Notes. This is not Sloan (1996) and not what Table A.1's own wording says ("changes in non-cash working capital minus depreciation"). It is nevertheless unambiguously what the paper's data does: netting depreciation out gives agreement 0.286, adding it gives 0.963, and decile one's share of market capitalisation comes out at 5.29% against a published 5.30%. Chen and Zimmermann's PctAcc.do carries the identical bracket placement while their Accruals.do does not, which suggests a shared transcription.

WORTH RAISING WITH THE COAUTHORS: as published, OA and AOA are not accruals net of depreciation, so capital-intensive firms sort toward the high-accrual leg because of their depreciation rather than away from it.

Rejected alternatives.

Variant	Agreement	Verdict
Sloan's grouping with dp subtracted outside the bracket	0.286	rejected
literal reading of A.1, all three current-liability terms subtracted	0.065	rejected
dp subtracted, scaled by average rather than lagged total assets	0.251	rejected
dp subtracted, every delta zero-filled	0.163	rejected
change in net operating assets	0.234	rejected
change in working capital from wcap directly	-0.058	rejected
cash-flow statement form, ni less oancf, over average assets	0.087	rejected
cash-flow form with balance-sheet fallback	0.269	rejected
scaled by sales, by absolute earnings, by current assets	0.154	rejected
winsorised 1/99 monthly, or —OA— capped at 1, 0.5, 0.3	0.065	no effect; outliers are not the cause
JKP oaccruals_at taken directly from their library	0.278	better than ours was, worse than the fix
shifted +/- 6 months against the published series	–	no peak; not a timing offset

AOA — Absolute operating accruals

Group intangibles **Source** Bandyopadhyay, Huang and Wirjanto (2010) **Frequency** annual **Agreement** 0.971

Inputs. `act, che, lct, dlc, txp, dp, at`

Formula.

AOA = —OA—, with OA exactly as above (depreciation added, not subtracted).

Notes. Follows OA mechanically. Neither Chen and Zimmermann nor Jensen, Kelly and Pedersen carry a Bandyopadhyay signal, so there is no third implementation to compare against; it is verified only through OA.

Rejected alternatives.

Variant	Agreement	Verdict
absolute value of every rejected OA variant above	0.087	rejected

DP — Dividend yield

Group value **Source** Litzenberger and Ramaswamy (1979) **Frequency** monthly **Agreement** 0.924

Inputs. `ret, retx, cap`

Formula.

dollars paid in month s = (ret[s] - retx[s]) * cap[s-1]
DP[t] = sum of those dollars over months t-11..t, divided by cap[t],
requiring at least six of the twelve months.

Notes. Dividends in dollars over market capitalisation, not dividends per share over price. Per share the share count drifts through the year and agreement falls to 0.841. JKP's div12m_me, which is this same quantity, scores 0.928 against the package, and building it natively gives 0.924. Against the paper's own panel (t_by_n/DP.mat) the two agree exactly where both are zero (53% of firm-months) but differ by more than 1% for three quarters of the non-zero ones, with the ratio spread from 0.87 to 1.10 rather than at any fixed factor, and the paper carries a value for 26% fewer firm-months than we do. No formula variant tried (below) narrows that, so the difference is in the dividend series itself, most plausibly dividends taken from CRSP's distribution events rather than from ret - retx. Verified at the sort level; the short leg is the widest gap among verified characteristics (4 points).

Rejected alternatives.

Variant	Agreement	Verdict
dividends per share over price (twelve-month sum)	0.841	rejected
per-share dividends restated for splits before summing	0.735	rejected; the more coherent calculation agrees worse
sum of twelve monthly dividend yields, no price	0.743	rejected
cash distributions from crsp.stkdistributions over price	0.348	rejected despite being the more principled source
dollar dividends over lagged rather than current market cap	0.869	rejected
JKP div_me (one month), div6m_me, div_at	0.652	rejected
dollar dividends over the previous month's market capitalisation	–	within 1% of the panel for 9% of non-zero firm-months against 25% for the current form
requiring all twelve months, or any one, instead of six	–	25% within 1% either way; no change
per-share dividends over the month-end price	–	28% within 1%, rank correlation 0.973 against 0.976; not better

ATO — Asset turnover

Group profitability **Source** Soliman (2008) **Frequency** annual **Agreement** 0.974
Inputs. `sale, at, che, ivao, dlc, dltt, mib, pstk, ceq`

Formula.

operating assets = at - che - ivao (che, ivao missing -¿ 0)
operating liabilities = at - dlc - dltt - mib - pstk - ceq
(dlc, dltt, mib, pstk missing -¿ 0; ceq REQUIRED)
net operating assets = operating assets - operating liabilities
ATO = sale / NOA[t-1]

Notes. Common equity must be required rather than zero-filled. Filled, operating liabilities collapse to assets less debt, net operating assets go to nearly zero, and anything divided by them explodes into the end deciles. Agreement 0.78 filled against 0.92 required. This is Chen and Zimmermann's convention in RetNOA.do, where dlc, dltt, mib and pstk are zero-filled and ceq is not.

Rejected alternatives.

Variant	Agreement	Verdict
ceq zero-filled like the other liability terms	0.780	rejected
contemporaneous rather than lagged net operating assets	0.582	rejected
operating assets without subtracting ivao	0.558	rejected
observations with non-positive NOA discarded	0.494	rejected
Soliman's component-built operating assets (rect+invt+aco+ppent+intan-ap-lco-lo), averaged	0.241	rejected
lagged sales over lagged NOA	0.625	rejected

RNA — Return on net operating assets

Group profitability **Source** Soliman (2008) **Frequency** annual **Agreement** 0.978
Inputs. `oiadp, at, che, ivao, dlc, dltt, mib, pstk, ceq`

Formula.

RNA = oiadp / NOA[t-1], with NOA exactly as defined under ATO
(ceq required, the other liability terms zero-filled).

Notes. Same common-equity point as ATO: 0.76 with ceq filled, 0.95 required.

Rejected alternatives.

Variant	Agreement	Verdict
ceq zero-filled	0.758	rejected
contemporaneous NOA	0.690	rejected
operating assets without ivao	0.551	rejected
non-positive NOA discarded	0.538	rejected
Soliman component-built operating assets	0.411	rejected

dSOUT — Change in shares outstanding

Group investment **Source** Pontiff and Woodgate (2008) **Frequency** monthly **Agreement** 0.992
Inputs. `shrout, mthcumfacshr`

Formula.

```
adjusted shares s[t] = shrout[t] * mthcumfacshr[t]
dSOUT[t] = 100 * ( s[t] / s[t-12] - 1 )
```

Notes. CRSP's share counts are not split-adjusted, so raw growth reads every split as a 100% issuance. The cumulative share factor is the exact adjustment.

Rejected alternatives.

Variant	Agreement	Verdict
raw shrout growth, unadjusted for splits	0.513	rejected
split adjustment backed out of market cap and the ex-dividend return	0.883	close, superseded

dPIA — Change in plant and inventory over assets

Group investment **Source** Lyandres, Sun and Zhang (2008) **Frequency** annual **Agreement** 0.943

Inputs. `ppegt, invt, at`

Formula.

```
dPIA = ( d ppegt + d invt ) / at[t-1],
where each delta treats a missing component as no change (zero) rather than
discarding the firm.
```

Rejected alternatives.

Variant	Agreement	Verdict
both deltas required non-missing	0.777	rejected
scaled by average rather than lagged assets	0.771	rejected
net rather than gross plant (ppent)	0.493	rejected
scaled by lagged (ppegt + invt)	0.420	rejected

TNOVR — Share turnover

Group trading frictions **Source** Datar, Naik and Radcliffe (1998) **Frequency** monthly **Agreement** 0.993

Inputs. `vol, shrout`

Formula.

TNOVR[t] = vol[t-1] / shrout[t]
the previous month's share volume over this month's shares outstanding, both from the CRSP monthly file.

Notes. RESOLVED against the paper's panel: Spearman 1.0000; agreement with the published decile weights 0.978 (a three-month average gave 0.79). The one-month lag of volume against the share count is what the data does; Table A.1 ("volume over shares outstanding") does not mention it and no reason for it suggests itself. Recorded under discrepancies, category B.

Rejected alternatives.

Variant	Agreement	Verdict
vol[t]/shrout[t], as documented	0.707	rejected
three-month average of vol/shrout	0.790	rejected; was the best available before the panel
twelve-month average	0.699	rejected
mean daily turnover within the month	0.706	rejected
Nasdaq volume scaled down 50% pre-1997 and 38% after	0.679	rejected
dollar volume over market cap	0.708	rejected
JKP turnover_126d	0.713	rejected

SUV — Standardised unexplained volume

Group trading frictions **Source** Garfinkel (2009), per Appendix Table A.1 **Frequency** monthly, sampled on the last trading day (daily inputs) **Agreement** 0.982

Inputs. `dlyvol, dlyret`

Formula.

On the trading days d of month t, the last day L included, fit by least squares
v_d = a + b * max(r_d, 0) + c * max(-r_d, 0) + e_d
with v_d the day's share volume and r_d its return. Then
SUV[t] = e_L / sqrt(sum_d e_d^2 / (n - 1)),
n the number of trading days in the month, at least 5.

Notes. RESOLVED against the paper's own panel (t_by_n/SUV.mat): Spearman 0.9999, Pearson 0.9998, and the ratio of the panel's values to ours is 1.0000 at every quantile (n - 3 in the denominator gives 1.054, n gives 0.976; what remains is float32 rounding in our daily file). Agreement with the published decile weights 0.982; decile one 8.94% against a published 8.94%. This is not Garfinkel's measure and not what Table A.1 describes: there the regression is estimated on the previous month and the surprise is the current month's volume; in the data the regression is estimated on the current month and the surprise is the last trading day's. Recorded under discrepancies, category A.

Rejected alternatives.

Variant	Agreement	Verdict
regression in volume levels rather than logs	0.080	rejected
estimation window of 1, 6, 12 and 24 months applied out of sample	0.092	rejected at every length
previous month taken on a dense grid so the lag cannot cross a gap	0.072	rejected
in-sample residual dispersion rather than an out-of-sample residual	0.002	rejected
standardised by sqrt(n) rather than n	0.073	rejected
documented reading at the daily level: month's mean (or sum over sqrt n) of daily standardised residuals from a trailing 63, 126 or 252-day fit, in volume, turnover or logs	–	Spearman against the panel between -0.001 and -0.04; rejected
last day's residual from a trailing window of 21, 42, 63, 90 days including the day	–	Spearman 0.990, 0.868, 0.809, 0.768: shorter is better, which pointed at the calendar month
trailing 63, 126, 252 days ending the day before	–	Spearman 0.803, 0.731, 0.674; rejected
plain z-score of the last day's volume against a trailing window, no regression	–	Spearman 0.75; rejected
log(1 + volume) or turnover as the dependent variable, within the month	–	Spearman 0.902 and 0.998 against 1.000 for share volume

DTO — Detrended turnover

Group trading frictions **Source** Garfinkel (2009), per Appendix Table A.1 **Frequency** monthly, sampled on the last trading day (daily inputs) **Agreement** 0.985

Inputs. `dlyvol, shrout (daily)`

Formula.

turn_d = dlyvol_d / shrout_d (daily share turnover)
DTO[t] = turn_L - mean(turn_d : d in the 180 trading days before L)
L is the last trading day of month t; the mean needs at least 90 of the 180 days.
No market turnover is subtracted, no Nasdaq volume scaling, no median, and no averaging over the month: the value on one day is the month's signal.

Notes. RESOLVED against the paper's own panel (t_by_n/DTO.mat, shared by Andrea Tamoni in September 2026): Spearman 0.998, Pearson 0.997 (the panel's values are ours times 500, a units convention that leaves every rank unchanged); agreement with the published decile weights 0.985, decile one 6.42% of market capitalisation against a published 6.50%. This is not what Table A.1 describes – "daily volume to shares outstanding minus the daily market turnover and detrended by the 180 trading day median", with Nasdaq volume scaled down – and each documented element lowers the agreement (below). Recorded under discrepancies, category A.

Rejected alternatives.

Variant	Agreement	Verdict
documented form, the 180-day median approximated by a rolling median of monthly means	0.367	rejected; rank correlation with the panel 0.28
last day, turnover less its 180-day median (120 and 252 days similar)	–	Spearman 0.888 against the panel; the mean gives 0.998
last day, (turnover - median)/median, and log turnover less log median	–	Spearman 0.78 and 0.80; rejected
subtracting the day's market turnover, and scaling Nasdaq volume by 0.50/0.38 (last day, median-detrended)	–	Spearman against the panel: neither 0.888; Nasdaq scaling alone 0.877; value-weighted market turnover subtracted 0.739, equal-weighted 0.666; both with Nasdaq scaling 0.643 and 0.564. Neither is in the data.
the month's mean of the daily series instead of its last day	–	rejected; the panel is the last day's value

aBEME — Industry-adjusted book to market

Group value **Source** Asness, Porter and Stevens (2000) **Frequency** monthly **Agreement** 0.896

Inputs. `beme, siccd (crsp.msenames name history)`

Formula.

> aBEME[t] = BEME[t] - mean(BEME[t] over ordinary common shares in the same industry that month)
> Equal-weighted over the firms that have BEME. The industry is the paper's own scheme: 27 groups of SIC codes recovered from its panels (raw/their_industry_table.json), the SIC code being the CRSP name-history code (crsp.msenames) in force that month. A code the table does not cover leaves aBEME missing (1.8% of firm-months).
> Recovery: within a month, X - aX is one constant across the firms of an industry; grouping SIC codes by that constant, month after month, gives 27 groups and no code-month pair that sits in two of them.

Notes. RESOLVED at 0.917 against the published decile weights once the industry scheme was recovered from the paper's panels (Fama-French 48 with the same inputs: see below). Table A.1 says the industries are Fama-French 48; the data's cells are 27 groups that align with FF48 only in part – one group holds 410 SIC codes spanning food, agriculture, textiles, clothing and construction, another 281 codes across building materials, machinery, steel, paper and household goods. The SIC source matters as well: the name-history code reproduces the panel's own siccd for 99.5% of firm-months. Recorded under discrepancies, category B.

Rejected alternatives.

Variant	Agreement	Verdict
Fama-French 48 with the same name-history SIC	–	rejected: 0.51 (aBEME), 0.45 (aSAT), 0.87 (aSIZE), 0.28 (aPM)
Fama-French 5, 10, 12, 17, 30, 38, 48 and 49 industries	0.521	ff30 marginally best, none close
one-digit, two-digit and three-digit SIC	0.477	rejected
Compustat historical SIC (sich), with and without CRSP fall-back	0.506	rejected
Compustat header SIC from comp.company	0.468	rejected
a firm's modal SIC over its life	0.509	rejected
industry median instead of mean	0.511	rejected
capitalisation-weighted industry mean	0.227	rejected
industry mean over all of CRSP rather than common shares only	0.099	rejected
industry mean over NYSE firms only	0.384	rejected
industry mean over the larger half of each industry	0.401	rejected
ratio to the industry mean, and log of that ratio	0.440	rejected
standardised within industry (divided by the industry sd)	0.348	rejected
percentile rank within industry	0.421	rejected
cross-sectional rank less the industry mean rank	0.391	rejected
leave-one-out industry mean	0.512	rejected
aggregate industry ratio (sum of book over sum of market)	0.226	rejected
partial adjustment, subtracting 0.25, 0.5 or 0.75 of the industry mean	0.458	rejected
previous year's industry mean	0.343	rejected

aSAT — Industry-adjusted asset turnover

Group profitability **Source** Soliman (2008) **Frequency** annual **Agreement** 0.896

Inputs. `sat, siccd (crsp.msenames name history)`

Formula.

aSAT[t] = SAT[t] - mean(SAT[t] over ordinary common shares in the same industry that month)
Equal-weighted over the firms that have SAT. The industry is the paper's own scheme: 27

groups of SIC codes recovered from its panels (raw/their_industry_table.json), the SIC code being the CRSP name-history code (crsp.msenames) in force that month. A code the table does not cover leaves aSAT missing (1.8% of firm-months).
Recovery: within a month, X - aX is one constant across the firms of an industry; grouping SIC codes by that constant, month after month, gives 27 groups and no code-month pair that sits in two of them.

Notes. RESOLVED at 0.906 against the published decile weights once the industry scheme was recovered from the paper's panels (Fama-French 48 with the same inputs: see below). Table A.1 says the industries are Fama-French 48; the data's cells are 27 groups that align with FF48 only in part – one group holds 410 SIC codes spanning food, agriculture, textiles, clothing and construction, another 281 codes across building materials, machinery, steel, paper and household goods. The SIC source matters as well: the name-history code reproduces the panel's own siccd for 99.5% of firm-months. Recorded under discrepancies, category B.

Rejected alternatives.

Variant	Agreement	Verdict
Fama-French 48 with the same name-history SIC	–	rejected: 0.51 (aBEME), 0.45 (aSAT), 0.87 (aSIZE), 0.28 (aPM)
the full classification and operation search listed under aBEME	0.448	ff48 demeaning best, none close
aggregate industry ratio (sum of sales over sum of assets)	0.264	rejected
partial adjustment at 0.25, 0.5, 0.75	0.261	rejected

aSIZE — Industry-adjusted size

Group trading frictions **Source** Asness, Porter and Stevens (2000) **Frequency** monthly **Agreement** 0.987

Inputs. `cap, siccd (crsp.msenames name history)`

Formula.

aSIZE[t] = SIZE[t] - mean(SIZE[t] over ordinary common shares in the same industry that month)
Equal-weighted over the firms that have SIZE. The industry is the paper's own scheme: 27 groups of SIC codes recovered from its panels (raw/their_industry_table.json), the SIC code being the CRSP name-history code (crsp.msenames) in force that month. A code the table does not cover leaves aSIZE missing (1.8% of firm-months).
Recovery: within a month, X - aX is one constant across the firms of an industry; grouping SIC codes by that constant, month after month, gives 27 groups and no code-month pair that sits in two of them.

Notes. RESOLVED at 0.992 against the published decile weights once the industry scheme was recovered from the paper's panels (Fama-French 48 with the same inputs: see below). Table A.1 says the industries are Fama-French 48; the data's cells are 27 groups that align with FF48 only in part – one group holds 410 SIC codes spanning food, agriculture, textiles, clothing and construction, another 281 codes across building materials, machinery, steel, paper and household goods. The SIC source matters as well: the name-history code reproduces the panel's own siccd for 99.5% of firm-months. Recorded under discrepancies, category B.

Rejected alternatives.

Variant	Agreement	Verdict
Fama-French 48 with the same name-history SIC	–	rejected: 0.51 (aBEME), 0.45 (aSAT), 0.87 (aSIZE), 0.28 (aPM)
Fama-French 10 and 30 industries	0.866	marginally better than 48, within noise
industry median instead of mean	0.877	marginally better, not adopted
log size demeaned within industry	0.754	rejected
partial adjustment at 0.75	0.872	marginally better, not adopted
capitalisation-weighted industry mean	0.317	rejected

A2ME — Total assets (at) over market capitalization (prc x shrout)

Group value **Source** bhandari1988 **Frequency** monthly **Agreement** 0.972

Inputs. `at`

Formula.

at / me.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December t-1	0.700	rejected

AT — Total assets (at)

Group trading frictions **Source** gandhi2015 **Frequency** annual **Agreement** 0.988

Inputs. `at`

Formula.

Total assets.

BEME — Book equity to market equity. Book equity is shareholders' equity (seq

Group value **Source** – **Frequency** monthly **Agreement** 0.977
Inputs. `at, ceq, lt, prc, pstk, pstkl, pstrkrv, seq, shrout, txditc`
Formula.

be / me, with negative book equity discarded before sorting.
Book equity = seq, else ceq+pstk, else at-lt; plus txditc; minus the first available of pstkrv, pstkl, pstk. Market equity is CRSP mthcap for the CURRENT month, in millions.

Notes. FNW state market equity as of December t-1 explicitly. The package's own stored panel says otherwise once its one-month row offset is accounted for: lined up correctly the current month matches 94.9% of firm-months exactly and December 3.9%. Implementing December drops agreement from 0.97 to about 0.70.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December of the prior year	0.700	rejected
market equity lagged one month	0.840	rejected; an artifact of a row-offset error in the comparison
market equity summed across share classes of the same permco	–	marginally better on values (95.7% vs 94.9%), not adopted

C2A — Cash and short-term investments (che) to total assets (at)

Group value **Source** palazzo2012 **Frequency** annual **Agreement** 0.963
Inputs. `at, che`
Formula.

che / at.

C2D — Cashflow to debt. Cashflow is the sum of income and extraordinary item

Group value **Source** ou1989 **Frequency** annual **Agreement** 0.959
Inputs. `dp, ib, lt`
Formula.

(ib + dp) / lt.

CAT — Sales (sale) to lagged total assets (at)

Group profitability **Source** haugen1996 **Frequency** annual **Agreement** 0.977
Inputs. `at, sale`
Formula.

sale / at[t-1].

D2P — Debt to price. Debt is long-term debt (dltt) plus debt in current liab

Group value **Source** litzenberger1979 **Frequency** monthly **Agreement** 0.964
Inputs. `dlc, dltt, prc, shrout`
Formula.

(dltt + dlc) / me, each debt term missing treated as zero.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December t-1	0.710	rejected

dCEQ — Annual % change in book value of equity (ceq)

Group investment **Source** richardson2005 **Frequency** annual **Agreement** 0.951
Inputs. `ceq`
Formula.

Percent change in ceq.

dGS — % change in gross margin minus % change in sales (sale). Gross margin

Group profitability **Source** abarbanell1997 **Frequency** annual **Agreement** 0.966
Inputs. `cogs, sale`
Formula.

Percent growth in gross margin (sale-cogs) less percent growth in sale.

dSO — Log change in the product of shares outstanding (csho) and the adjustm

Group value **Source** fama2008 **Frequency** annual **Agreement** 0.966
Inputs. `ajex, csho`
Formula.

log(csho*ajex / (csho*ajex)[t-1]), i.e. growth in split-adjusted Compustat share count.

E2P — Income before extraordinary items (ib) to market capitalization (prc x

Group value **Source** basu1983 **Frequency** monthly **Agreement** 0.978
Inputs. `ib`
Formula.

ib / me.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December t-1	0.690	rejected

EPS — Income before extraordinary items (ib) to shares outstanding (shrout)

Group value **Source** basu1977 **Frequency** monthly **Agreement** 0.977
Inputs. `ib, shrout`
Formula.

ib / (shrout/1000), i.e. earnings per share on CRSP's share count.

IDIOV — Standard deviation of the residuals from a regression of excess return

Group trading frictions **Source** ang2006 **Frequency** monthly **Agreement** 0.955
Formula.

Standard deviation of the residuals from a regression of daily excess returns on the three Fama-French factors, within the month, at least 15 days.
The residual standard deviation divides the residual sum of squares by n - 1, not by n - 4.

Notes. Emphatically monthly, unlike BETA_d: 0.93 at one month against 0.58 at twelve. JKP corroborate, carrying ivol_ff3_21d at a 21-day window, and our series correlates 0.997 with theirs. Against the paper's panel the version dividing by n - 4 ran 8.45% high at the median month; (ours/theirs)^2 by number of trading days is (n - 1)/(n - 4) to four decimals at every n from 15 to 23, so the panel is the sample standard deviation of the residuals. Ranks move very little (rank correlation was already 0.9997). Over the full panel: within 1% for 75.8% of firm-months, rank correlation 0.9998, median ratio 1.0000; the remainder is the factor vintage on small values.

Rejected alternatives.

Variant	Agreement	Verdict
estimated over twelve months	0.582	rejected
residual variance with n - 4 degrees of freedom	0.955	a convention, not an error; the panel divides by n - 1

I2A — Annual % change in total assets (at)

Group investment **Source** – **Frequency** annual **Agreement** 0.971
Inputs. `at`
Formula.

Percent change in at.

IPM — Pre-tax income (pi) over sales (sale)

Group profitability **Source** – **Frequency** annual **Agreement** 0.962
Inputs. `pi, sale`
Formula.

pi / sale.

Notes. A zero sales denominator gives a missing value, not an infinity: the paper's panel has no infinities, and an infinity inside an industry mean would wipe out the whole industry-month for aPM.

IVC — Annual change in inventories (invt) in the last two fiscal years over

Group investment **Source** thomas2002 **Frequency** annual **Agreement** 0.970
Inputs. `at, invt`
Formula.

(invt - invt[t-1]) / ((at + at[t-1])/2).

SIZE — Price (prc) times shares outstanding (shrout)

Group trading frictions **Source** fama1992 **Frequency** monthly **Agreement** 0.997
Inputs. `prc, shrout`
Formula.

Market equity, CRSP mthcap for the current month, in millions.

Notes. Verified at the firm level against the paper panel SIZE.mat in Persistence_BBOT/replicatePWsorts/DataIN: 96.3% of firm-months exact to 1e-6, 99.0% within 1%, rank correlation 0.9998. Current month, not lagged and not December, and in millions – CRSP thousands give a median ratio of exactly 1000.

NOA — Operating assets minus operating liabilities to lagged total assets (a

Group investment **Source** hirshleifer2004 **Frequency** annual **Agreement** 0.967
Inputs. `at, ceq, che, dlc, dltt, ivao, mib, pstk`
Formula.

> (operating assets - operating liabilities) / at[t-1], with operating assets
> and liabilities as defined under ATO.

Notes. Unaffected by the common-equity question that matters so much for ATO and RNA, because NOA uses the level rather than dividing by it.

Rejected alternatives.

Variant	Agreement	Verdict
ceq zero-filled	0.917	no effect

OL — Sum of cost of goods sold (cogs) and selling, general and administrati

Group intangibles **Source** novy2011 **Frequency** annual **Agreement** 0.959
Inputs. `at, cogs, xsga`
Formula.

> (cogs + xsga) / at, each missing term treated as zero.

BETA_d — The sum of the regression coefficients of daily excess returns on mark

Group trading frictions **Source** lewellen2006 **Frequency** monthly, sampled on the last trading day (daily inputs) **Agreement** 0.990
Inputs. `dlyret, ff.factors_daily mktrf and rf`
Formula.

> Over the firm's last 249 trading days ending on the month's last trading day, fit
> r_d - rf_d = a + b0 * mktrf_d + b1 * mktrf_{d-1} + e_d
> where mktrf_{d-1} is the market's value on the firm's previous trading day, and
> BETA_d[t] = b0 + b1.
> At least 10 of the firm's days in the window; a firm with a shorter history uses
> what it has.

Notes. RESOLVED against the paper's own panel (t_by_n/BETA_d.mat): 84% of firm-months within 1%, rank correlation 0.9996, median ratio 1.0000. The window is 249 of the firm's own trading rows, which is what a 250-day window gives once the lag consumes its first day; 248 or 250 rows halve the agreement (0.49 and 0.47), calendar twelve months gives 0.24, and a window counted on the market's calendar rather than the firm's rows 0.76. The firm-months outside 1% have betas below 0.2 in absolute value, where a relative criterion is harsh, and thin out from 45% in the first year of the daily sample to

5-18% after; the ranking is exact. Table A.1 states no window at all. Over the full panel: within 1% for 89.7% of firm-months, rank correlation 0.9994, median ratio 1.0002.

Rejected alternatives.

Variant	Agreement	Verdict
estimated over one month	0.353	rejected
six months	0.617	rejected
twenty-four and sixty months	0.631	rejected
calendar twelve months of daily data (the previous form)	0.952	superseded: 24% of firm-months within 1% of the panel
250, 251, 252, 248, 245 or 240 of the firm's trading days	–	47%, 35%, 28%, 49%, 26%, 13% within 1%; 249 gives 84%
250 days on the market calendar, thin traders having fewer observations	–	76% at 249 days, 43% at 250; the window counts the firm's own days
CRSP value-weighted or equal-weighted index instead of the Fama-French market factor	–	10% and 2% within 1%, ratio 1.03 and 1.17; rejected
window ending the day before the month's last trading day	–	28%; the window ends on the last day
plain market beta without the lag term	–	2% within 1%, ratio 0.80; the panel is the Dimson sum

PCM — Net sales (sale) minus cost of goods sold (cogs) all scaled by net sal

Group profitability **Source** gorodnichenko2016 **Frequency** annual **Agreement** 0.977
Inputs. `cogs, sale`

Formula.

(sale - cogs) / sale.

Notes. A zero sales denominator gives a missing value, not an infinity: the paper's panel has no infinities, and an infinity inside an industry mean would wipe out the whole industry-month for aPM.

PM — Operating Income after depreciation (oiadp) to sales (sale)

Group profitability **Source** soliman2008 **Frequency** annual **Agreement** 0.973
Inputs. `oiadp, sale`

Formula.

oiadp / sale.

Notes. A zero sales denominator gives a missing value, not an infinity: the paper's panel has no infinities, and an infinity inside an industry mean would wipe out the whole industry-month for aPM.

Rejected alternatives.

Variant	Agreement	Verdict
net income over revenue, as Chen and Zimmermann implement Soliman	0.387	rejected
oiadp over average sales	0.732	rejected

PROF — Gross profitability (gp) over book equity as defined in BEME

Group profitability **Source** ball2015 **Frequency** annual **Agreement** 0.978
Inputs. `gp`
Formula.

> gp / be, with book equity as defined under BEME.

Q — Total assets (at) plus market value of equity (shrout x prc) minus boo

Group value **Source** – **Frequency** monthly **Agreement** 0.978
Inputs. `at, ceq, txdb`
Formula.

> (at + me - ceq - txdb) / at, txdb missing treated as zero, me the current month's market equity in millions.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December of the prior year	0.610	rejected

MAXRET — Maximum daily return in the previous month

Group trading frictions **Source** bali2011 **Frequency** monthly **Agreement** 0.984
Formula.

> Maximum daily return within the month.

Notes. Correlates 1.000 with JKP's rmax1_21d.

ROA — Income before extraordinary items (ib) to lagged total assets (at)

Group profitability **Source** balakrishnan2010 **Frequency** annual **Agreement** 0.965

Inputs. `at, ib`

Formula.

ib / at[t-1].

ROC — Market value of equity (shrout x prc) plus long-term debt (dltt) minus

Group profitability **Source** chandrashekar2009 **Frequency** monthly **Agreement** 0.972
Inputs. `at, che, dltt`

Formula.

(me + dltt - at) / che.

ROE — Income before extraordinary items (ib) to lagged book-value of equity

Group profitability **Source** haugen1996 **Frequency** annual **Agreement** 0.972
Inputs. `ib`

Formula.

ib / be[t-1].

ROIC — Earnings before interest and taxes (ebit) less non-operating income (n

Group profitability **Source** brown2007 **Frequency** annual **Agreement** 0.964
Inputs. `ceq, che, ebit, lt, nopi`

Formula.

(ebit - nopi) / (ceq + lt - che), nopi missing treated as zero.

Notes. Appendix Table A.1 words the denominator as a sum, which would add cash to invested capital. Subtracting it is both what invested capital means and what the data says: 0.64 adding against 0.93 subtracting.

Rejected alternatives.

Variant	Agreement	Verdict
cash added to invested capital, as A.1 words it	0.640	rejected
every component required non-missing	0.640	no effect

R_12_2 — Cumulative return from 12 months to 2 months ago

Group past returns **Source** fama1996 **Frequency** monthly **Agreement** 0.979

Formula.

Compound return over months t-11..t-1, stored on row t.

Notes. Stored ONE MONTH EARLIER than the name suggests. The sort's own one-month lag then produces the stated twelve-to-two window. Storing the literal window drops agreement from 0.97 to 0.65. Verified at the firm level against the paper panel: rank correlation 0.9989. Theirs is stored as the gross return, 1 + r, so no value matches to the digit and the median ratio ours over theirs is 0.05; ranks are identical.

Rejected alternatives.

Variant	Agreement	Verdict
literal window, months t-12..t-2	0.647	rejected

R_12_7 — Cumulative return from 12 months to 7 months ago

Group past returns **Source** novy2012 **Frequency** monthly **Agreement** 0.984
Formula.

Compound return over months t-11..t-6, stored on row t.

Rejected alternatives.

Variant	Agreement	Verdict
literal window t-12..t-7	0.533	rejected

R_6_2 — Cumulative return from 6 months to 2 months ago

Group past returns **Source** jegadeesh1993 **Frequency** monthly **Agreement** 0.978
Formula.

Compound return over months t-5..t-1, stored on row t.

Rejected alternatives.

Variant	Agreement	Verdict
literal window t-6..t-2	0.507	rejected

R_2_1 — Lagged one month return

Group past returns **Source** jegadeesh1990 **Frequency** monthly **Agreement** 0.986
Formula.

The CURRENT month's return, stored on row t.

Notes. The one-month shift matters most here: with a single month there is no overlap to rescue a timing error, and the literal 'lagged one-month return' gives 0.11 against 0.97.

Rejected alternatives.

Variant	Agreement	Verdict
lagged one-month return, as A.1 words it	0.108	rejected

R_36_13 — Cumulative return from 36 months to 13 months ago

Group past returns **Source** de1985 **Frequency** monthly **Agreement** 0.988

Formula.

Compound return over months t-35..t-12, stored on row t.

Rejected alternatives.

Variant	Agreement	Verdict
literal window t-36..t-13	0.744	rejected

S2C — Net sales (sale) to cash and short-term investments (che)

Group value **Source** ou1989 **Frequency** annual **Agreement** 0.968

Inputs. `che`, `sale`

Formula.

sale / che.

S2P — Net sales (sale) to market capitalization (shrout x prc)

Group value **Source** lewellen2015 **Frequency** monthly **Agreement** 0.979

Inputs. `sale`

Formula.

sale / me.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December	0.720	rejected

SG — % growth rate in sales (sale)

Group value **Source** lakonishok1994 **Frequency** annual **Agreement** 0.963
Inputs. `sale`

Formula.

Percent change in sale.

SAT — Sales (sale) to total assets (at)

Group profitability **Source** soliman2008 **Frequency** annual **Agreement** 0.971
Inputs. `at, sale`

Formula.

sale / at.

Rejected alternatives.

Variant	Agreement	Verdict
sale over average assets	0.786	rejected
sale over lagged assets	0.736	rejected

SPREAD — Average daily bid-ask spread in the previous month

Group trading frictions **Source** Chung and Zhang (2014), per Appendix Table A.1 **Frequency** monthly **Agreement** 0.985
Inputs. `dlyhigh, dlylow, dlybid, dlyask`

Formula.

Mean over the month's trading days of a daily spread, where each day contributes
2 (high - low) / (high + low) if the day has a high and a low,
2 (ask - bid) / (ask + bid) otherwise, if it has quotes,
and days with neither are left out. No minimum number of days.

Notes. RESOLVED against the paper's own panel (t_by_n/SPREAD.mat): built day by day this way, every firm-month in 1975, 1985, 1995 and 2005 agrees with the panel within 1%, and no firm-month the panel has is missing. FNW cite Chung and Zhang (2014), whose estimator is the quoted spread; CRSP carries quotes for 9% of daily rows in the 1960s and 53% in the 1980s, so the quoted spread alone scores 0.02 against the published decile weights, and the range alone 0.94 but with all of 1970s Nasdaq missing. The month-level fallback that preceded this (range when fifteen days have it, else quotes) reached 0.973 on the published weights but agreed with the panel for only 69-88% of firm-months. Over the full 1960-2017 panel the rebuilt series matches the paper's exactly for 99.4% of firm-months and within 1% for 100%.

Rejected alternatives.

Variant	Agreement	Verdict
quoted bid-ask spread, 2*(ask-bid)/(ask+bid)	0.017	rejected
high-low range over the closing price	0.933	marginally worse
range when at least 15 days have one, else the quoted spread when at least 15 days have quotes	0.973	superseded: within 1% of the panel for 69-88% of firm-months by year
quoted spread first, range on the days without quotes	–	rejected: within 1% for 0.2-100% by year; the range comes first
day-level mixed rule with a 15- or 10-day minimum	–	same values; leaves 2-139 firm-months a year missing that the panel has

sdTURN — Standard deviation of residuals from a regression of daily turnover on

Group trading frictions **Source** chordia2001 **Frequency** monthly **Agreement** 0.991
Inputs. `shrout`, `vol`

Formula.

> Standard deviation within the month of daily turnover (volume over shares outstanding), requiring at least 15 days.

Rejected alternatives.

Variant	Agreement	Verdict
coefficient of variation of turnover (Chordia et al.)	0.004	rejected

sdDVOL — Standard deviation of residuals from a regression of daily volume (vol

Group trading frictions **Source** chordia2001 **Frequency** monthly **Agreement** 0.989
Inputs. `vol`

Formula.

> Standard deviation (n - 1) over the month's trading days that have volume of
> ln(dollar volume_d), dollar volume = CRSP dlyprcvol,
> days with zero volume left out; at least 15 trading days in the month and at least 2 with volume.

Notes. In LOGS. In levels the dispersion measures scale rather than variability: 0.06 against 0.87. Against the paper's own panel (t_by_n/sdDVOL.mat) the earlier ln(1 + dollar volume) over all days matched for the 73% of firm-months without a zero-volume day and disagreed by a factor of 1.6 to 5 for every firm-month with one: a zero-volume day enters as ln(1) = 0 and dominates the month's dispersion. Leaving those days out, in ln(x) rather than ln(1 + x), is what the panel does. Over the full panel: exact for 90.9% of firm-months, within 1% for 100%, rank correlation 1.0000.

Rejected alternatives.

Variant	Agreement	Verdict
standard deviation of dollar volume in levels	0.057	rejected
standard deviation of log share volume	0.632	rejected
coefficient of variation of dollar volume (Chordia et al.)	0.521	rejected
ln(1 + dollar volume) over all trading days, zero-volume days included	0.896	rejected once the panel showed the zero-volume days are excluded; rank correlation with the panel 0.888

TAN — Tangibility is defined as (0.715 x total receivables (rect) + 0.547 x

Group intangibles **Source** hahn2009 **Frequency** annual **Agreement** 0.978
Inputs. `at, che, invt, ppent, rect`
Formula.

(0.715*rect + 0.547*invt + 0.535*ppent + che) / at, every component REQUIRED.

Notes. Zero-filling a missing component reads as 'this firm holds no tangible assets' and sends firms with incomplete filings to the bottom decile: 0.48 against 0.96.

Rejected alternatives.

Variant	Agreement	Verdict
missing components filled with zero	0.478	rejected

RETVOL — Standard deviation of residuals from a regression of excess returns on

Group trading frictions **Source** ang2006 **Frequency** monthly **Agreement** 0.988
Formula.

Standard deviation of daily excess returns within the month, at least 15 days.

Notes. Correlates 1.000 with JKP's rvol_21d.

aPM — Industry-adjusted profit margin

Group profitability **Source** Soliman (2008) **Frequency** annual **Agreement** 0.778
Inputs. `pm, siccd (crsp.msenames name history)`
Formula.

> aPM[t] = PM[t] - mean(PM[t] over ordinary common shares in the same industry that month)
> Equal-weighted over the firms that have PM. The industry is the paper's own scheme: 27 groups of SIC codes recovered from its panels (raw/their_industry_table.json), the SIC code being the CRSP name-history code (crsp.msenames) in force that month. A code the table does not cover leaves aPM missing (1.8% of firm-months).
> Recovery: within a month, X - aX is one constant across the firms of an industry; grouping SIC codes by that constant, month after month, gives 27 groups and no code-month pair that sits in two of them.

Notes. RESOLVED against the published decile weights once the industry scheme was recovered from the paper's panels (Fama-French 48 with the same inputs: see below). Table A.1 says the industries are Fama-French 48; the data's cells are 27 groups that align with FF48 only in part – one group holds 410 SIC codes spanning food, agriculture, textiles, clothing and construction, another 281 codes across building materials, machinery, steel, paper and household goods. The SIC source matters as well: the name-history code reproduces the panel's own siccd for 99.5% of firm-months. Its parent PM is missing, not infinite, when sales are zero; an infinity inside an industry mean wiped out every firm in that industry-month, which is what had kept aPM at 0.37 after the scheme was recovered. Recorded under discrepancies, category B.

Rejected alternatives.

Variant	Agreement	Verdict
Fama-French 48 with the same name-history SIC	–	rejected: 0.51 (aBEME), 0.45 (aSAT), 0.87 (aSIZE), 0.28 (aPM)
the full classification and operation search listed under aBEME	0.340	two-digit SIC marginally best, none close
partial adjustment at 0.25, 0.5, 0.75	0.261	rejected
industry mean over the larger half, and the prior year's mean	0.146	rejected